If/Then Decisions

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

WORKBOOK · 1 OF 21

Recall and choose your starting point

> Record your starting explanation for If/Then Decisions before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Algorithms as Recipes

Without opening the old lesson, explain one key idea from Algorithms as Recipes. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Categories and Labels

Without opening the old lesson, explain one key idea from Categories and Labels. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Rules or Learned Patterns?

Without opening the old lesson, explain one key idea from Rules or Learned Patterns?. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported If/Then Decisions trace and record the first step you can explain without help.

Use W2: Read four boundary phrases for **Condition**s and boundary cases. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed If/Then Decisions explanation against the supplied facts before discussing it.

Use the worked case for **Condition**s and boundary cases. Proposed explanation: “The **condition** reads the starting value”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Transfer the method to a new display

A paper brightness display adds 3 to its starting number. If the current brightness is at most 7, subtract 2; otherwise add 1. Report and stop. Trace separate runs from starting values 3, 4 and 5.

Identify the run that tests equality. Then show how its result changes if at most 7 is accidentally written as below 7.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

Connect the If/Then Decisions target to a trace and a justified conclusion.

A club admits a visitor if they have a ticket AND an accompanying adult. Test all four yes/no combinations. A student changes AND to OR. Which outcomes change?

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied If/Then Decisions case and show the deciding evidence.

If a card has a star, send it to A; otherwise send it to B.

Sort a star, a circle, and a card with both shapes.

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

Use this exact rule: if a card has more than three dots, place it in tray A; otherwise use tray B. Trace cards with 2, 3 and 4 dots.

My independent answer and reason

Write or draw your reasoning and evidence.

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

WORKBOOK · 9 OF 21

Project checkpoint

> Justify what If/Then Decisions adds to Helpful Classroom Sorter and keep the evidence.

Add one transparent if/then **branch** and one **else** path to the Classroom Sorter.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

A club admits a visitor if they have a ticket AND an accompanying adult. Test all four yes/no combinations. A student changes AND to OR. Which outcomes change?

My independently reasoned answer

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

W1 Trace a decision from the current score

Trace a decision from the current score. Apply the supplied If/Then Decisions facts and keep your reasoning.

Use the paper-game procedure: add 2 to score; if score is at least 6 subtract 3, otherwise add 4; report and stop. Start at 2, then make a separate run starting at 5.

For each run record the value at the **condition**, its true or false result, the **branch** and the final score.

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Read four boundary phrases

> Read four boundary phrases. Apply the supplied If/Then Decisions facts and keep your reasoning.

The known value at a **condition** is exactly 6. Decide whether each test is true or false: at least 6; above 6; at most 6; below 6. Explain which phrases include equality.

Then name one value below and one value above 6 for neighbouring tests.

My reasoning and evidence

WORKBOOK · 14 OF 21

W3 Stop after the selected branch

> Stop after the selected branch. Apply the supplied If/Then Decisions facts and keep your reasoning.

In a separate decision the current score is 4. If score is at least 5 subtract 2; otherwise add 3; report and stop. A learner adds 3 to get 7, then subtracts 2 because 7 is now at least 5, and reports 5.

Identify the extra action and give the correct trace and result.

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Transfer the method to a new display

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Transfer the method to a new display. Apply the supplied If/Then Decisions facts and keep your reasoning.

A paper brightness display adds 3 to its starting number. If the current brightness is at most 7, subtract 2; otherwise add 1. Report and stop. Trace separate runs from starting values 3, 4 and 5.

Identify the run that tests equality. Then show how its result changes if at most 7 is accidentally written as below 7.

My reasoning and evidence

WORKBOOK · 16 OF 21

W5 Handle an unknown input explicitly

> Handle an unknown input explicitly. Apply the supplied If/Then Decisions facts and keep your

> reasoning.

The score is unreadable at the at-least-6 test. A helper says the test must be false and takes the otherwise **branch**. Explain what is unknown and why that choice is unjustified.

State the next step required by this paper exercise and give two possible scores that would select different **branch**es.

> **My reasoning and evidence**

WORKBOOK · 17 OF 21

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

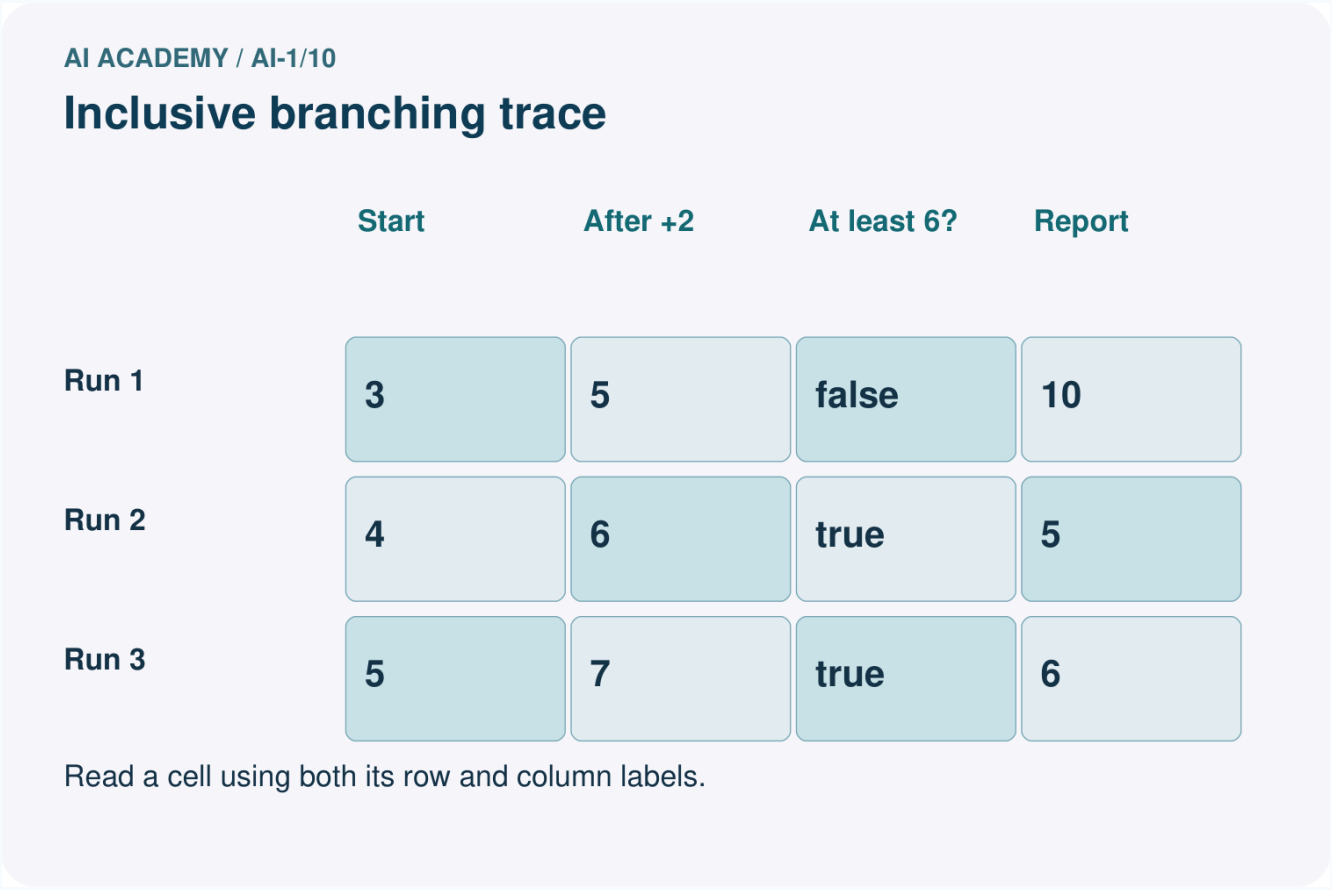
Pictorial case practice: If/Then Decisions

> A fictional counter begins with a number, adds 2, then chooses one branch. Explain the labelled step before checking the changed condition.

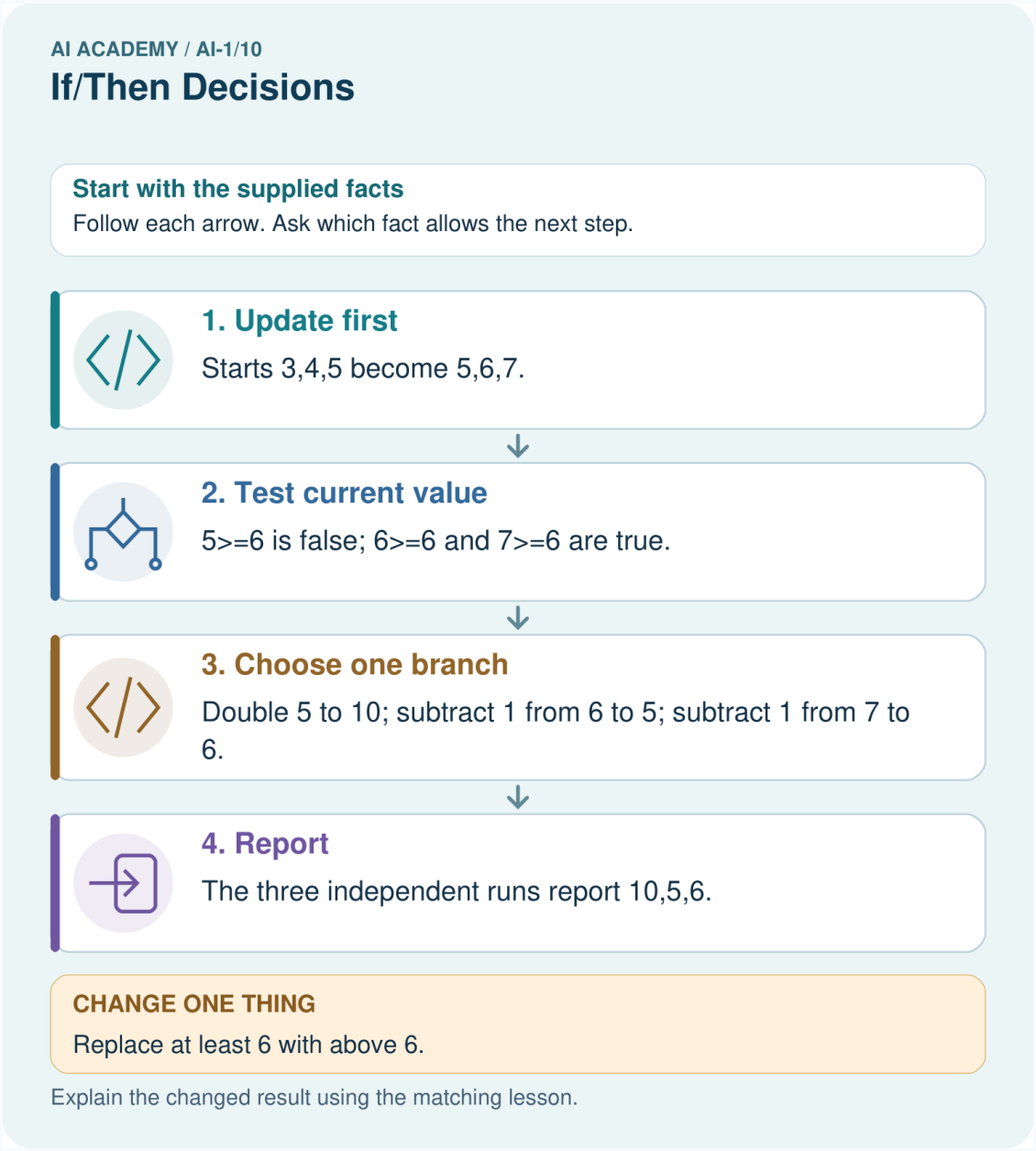
A fictional counter begins with a number, adds 2, then chooses one **branch**.

If the updated value is at least 6, subtract 1; otherwise double it.

Report once and stop. Use separate starting values 3,4,5.



Follow the labelled picture with the matching explanation. Open the vector diagram



Follow the labelled picture with the matching explanation. Open the vector diagram

Which **input** detects the difference between the two written **condition**s? Explain its entire run.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Include equality examples when reviewing sorter **condition**s, and trace from original **input** through **output**.

WORKBOOK · 19 OF 21

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Helpful Classroom Sorter record.

Project: Helpful Classroom Sorter

Current milestone: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

Include equality examples when reviewing sorter **condition**s, and trace from original **input** through **output**.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional counter begins with a number, adds 2, then chooses one **branch**.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? The three independent runs report 10,5,6.

1. Update first

2. Report

3. Choose one **branch**

4. Test current value

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Report The three independent runs report 10,5,6. The **condition** is evaluated after adding 2.

Equality belongs to the at-least **branch**, so starting at 4 checks the boundary. Once a **branch** finishes, the program reports; it does not reconsider the **condition** just because doubling 5 produced 10.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Replace at least 6 with above 6. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. Only the run starting at 4 changes: it reports 12.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed **condition** and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Only the run starting at 4 changes: it reports 12. Updated value 6 fails a strict above comparison and is doubled.

Repair a missing building block

Review Algorithms as Recipes (ai-1/9). Goal: Trace an ordered algorithm using the current value after every instruction. Short practice: Use the arrows to explain where the two zero-start runs first differ..

Return to this week and explain what you changed.

Review Categories and Labels (ai-1/7). Goal: Apply a category definition using explicit **evidence** and boundary rules. Short practice: Explain why B originally was revise rather than review, then label its changed version..

Return to this week and explain what you changed.

Review Rules or Learned Patterns? (ai-1/4). Goal: Execute a stated rule including its boundary. Short practice: Show which stage used labels and which stage used only the new measurement..

Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

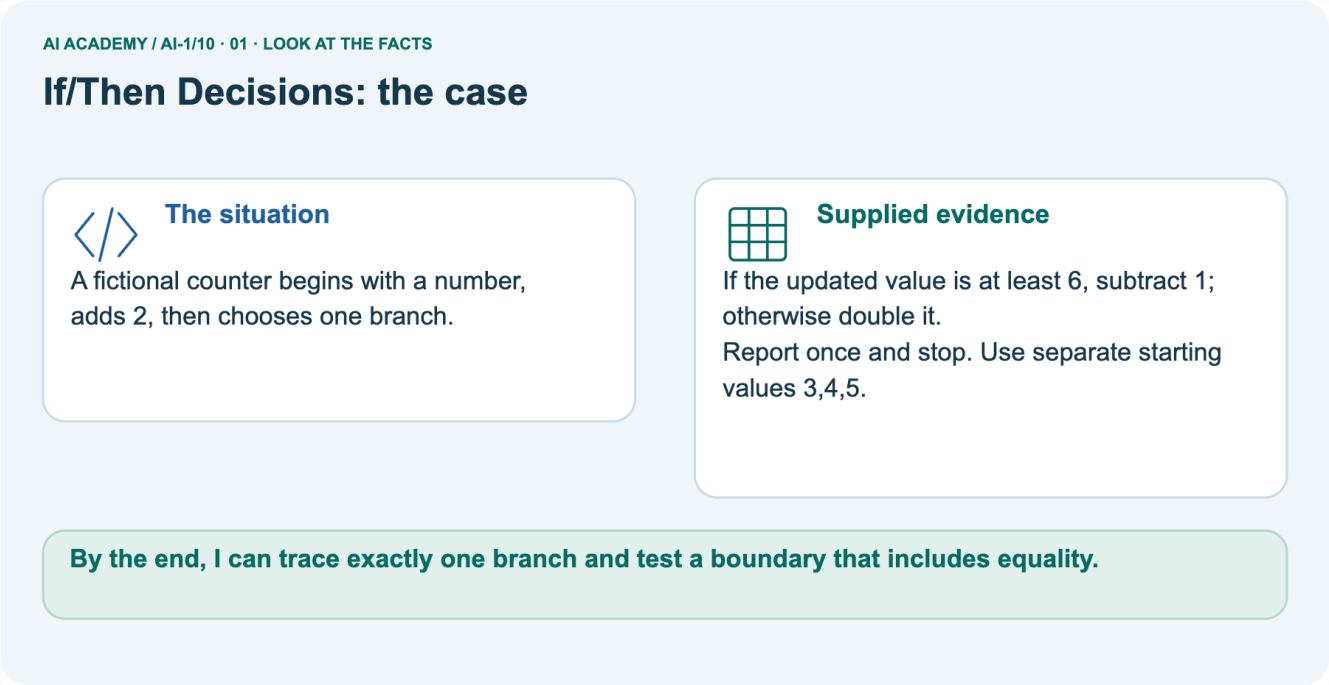
> **What I repaired, the evidence I kept and the next test**
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> | |
> |---|
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WORKBOOK · 21 OF 21

Visual concept journey

> Explain each If/Then Decisions picture from its labels, then record a relevant limit.

01. If/Then Decisions: the case



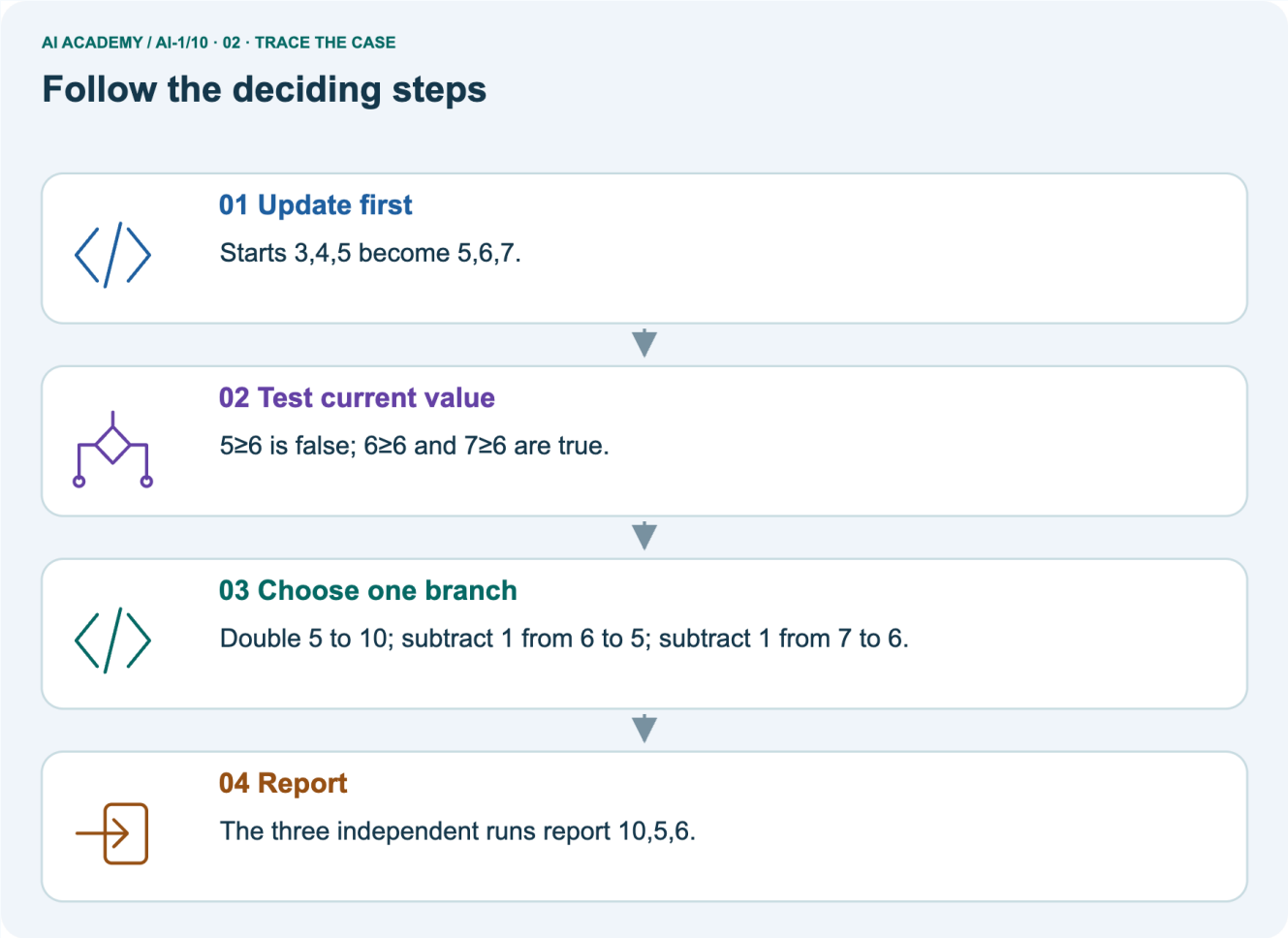
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional counter begins with a number, adds 2, then chooses one **branch**. Supplied facts: If the updated value is at least 6, subtract 1; otherwise double it. Report once and stop. Use separate starting values 3,4,5.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



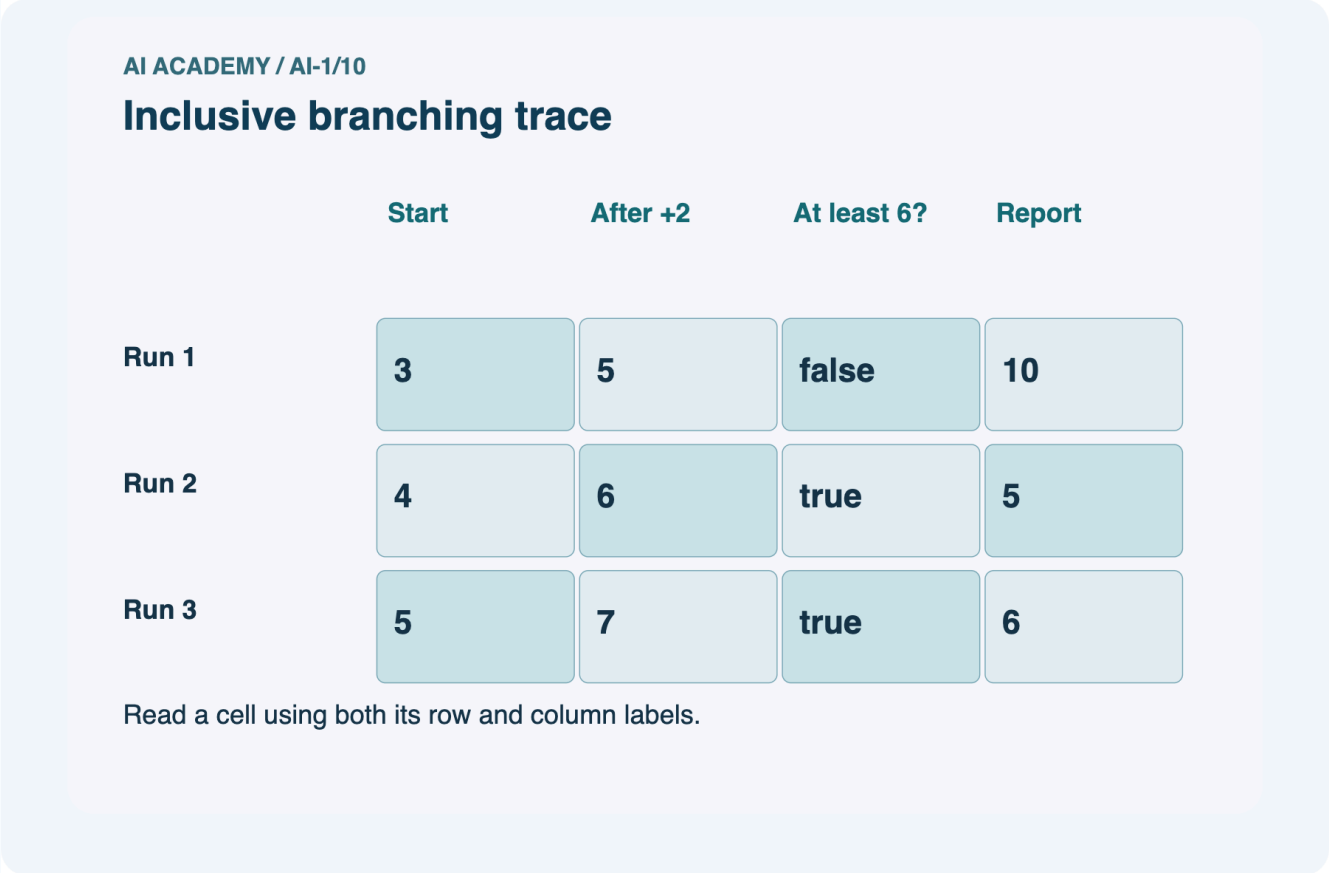
Follow the labelled picture with the matching explanation. Open the vector diagram

Update first: Starts 3,4,5 become 5,6,7. → Test current value: 5≥6 is false; 6≥6 and 7≥6 are true. → Choose one **branch**: Double 5 to 10; subtract 1 from 6 to 5; subtract 1 from 7 to 6. → Report: The three independent runs report 10,5,6.

> Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Inclusive branching trace



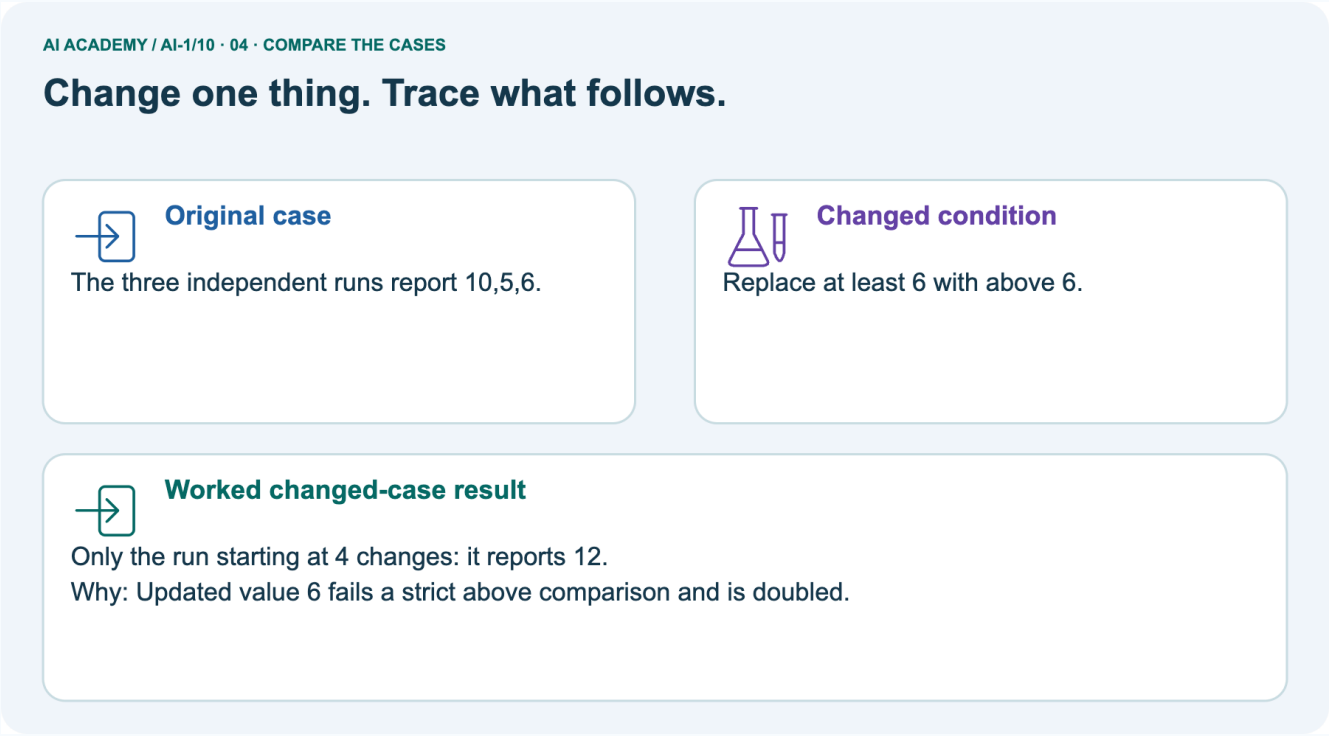
Follow the labelled picture with the matching explanation. Open the vector diagram

Inclusive **branch**ing trace. Run 1: Start = 3, After +2 = 5, At least 6? = false, Report = 10. Run 2: Start = 4, After +2 = 6, At least 6? = true, Report = 5. Run 3: Start = 5, After +2 = 7, At least 6? = true, Report = 6.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

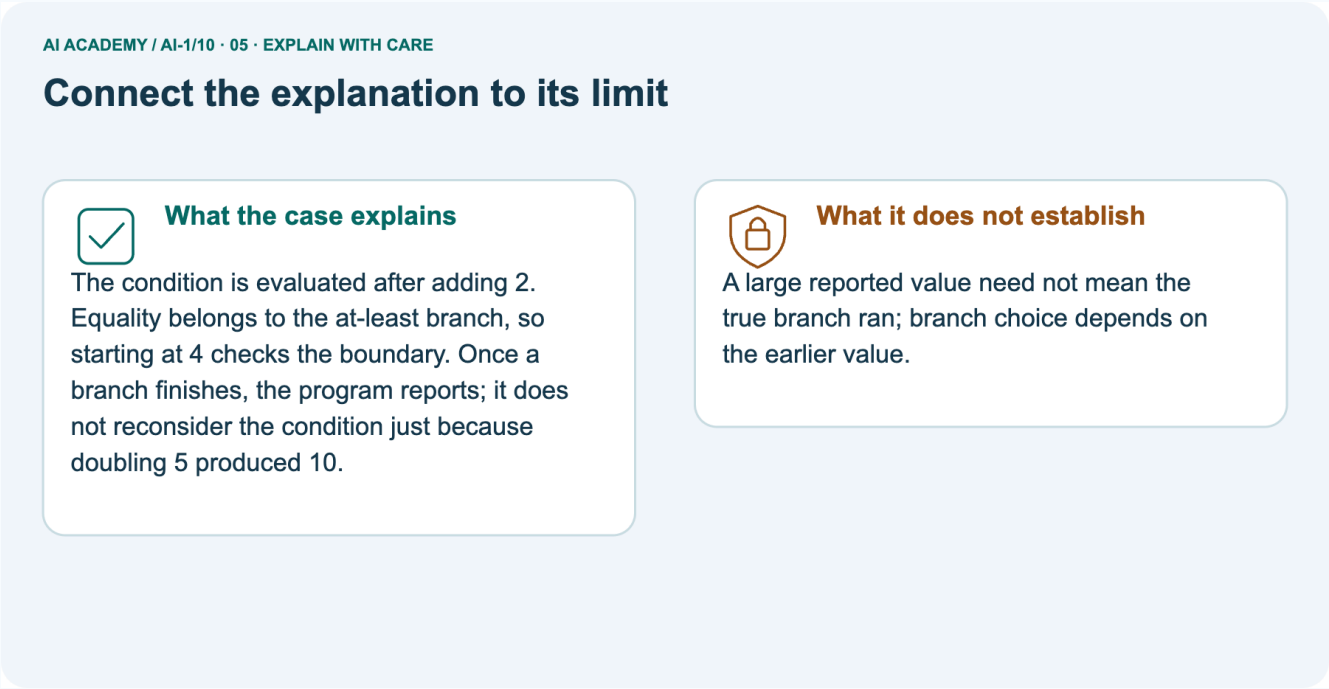
Original case: The three independent runs report 10,5,6. Changed **condition**: Replace at least 6 with above 6. Expected result: Only the run starting at 4 changes: it reports 12.

Reason: Updated value 6 fails a strict above comparison and is doubled.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

The **condition** is evaluated after adding 2. Equality belongs to the at-least **branch**, so starting at 4 checks the boundary.

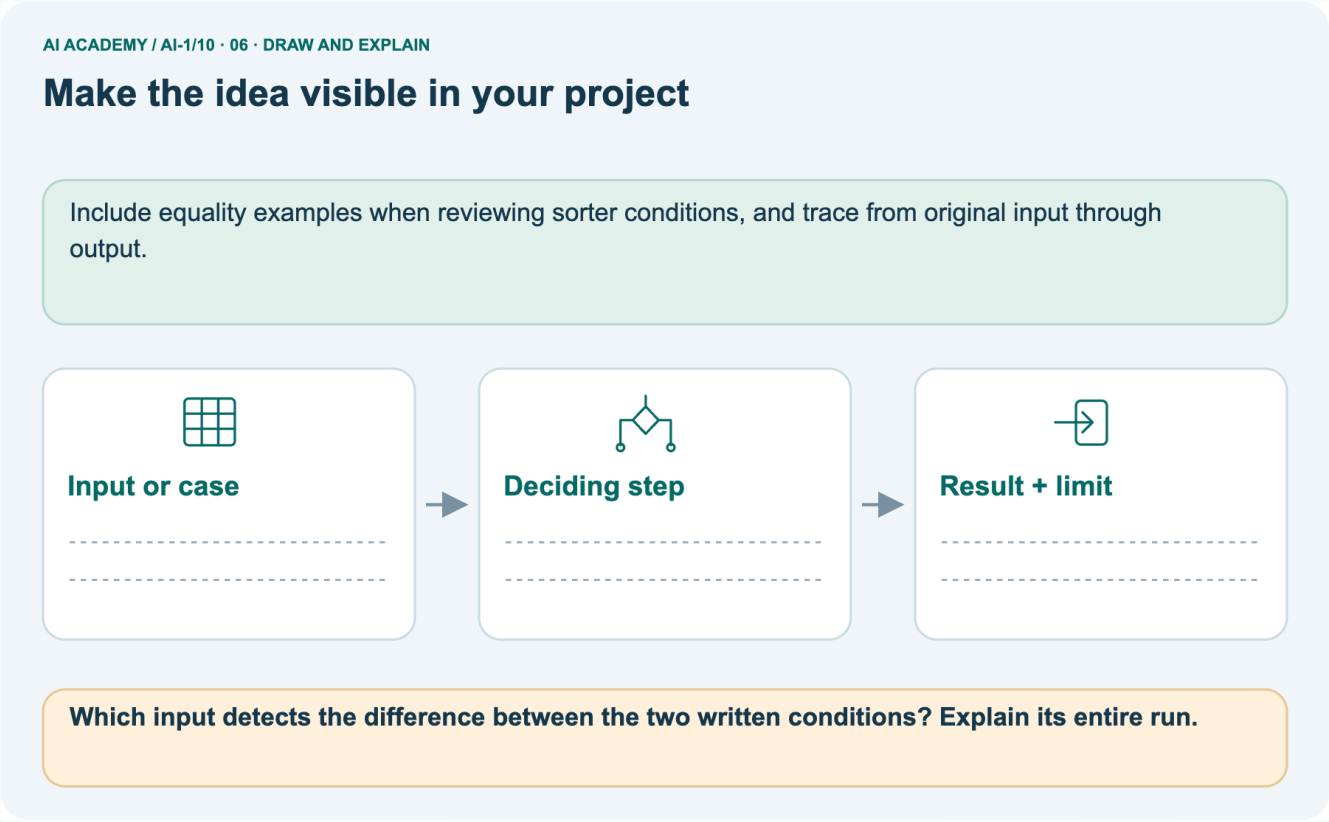
Once a **branch** finishes, the program reports; it does not reconsider the **condition** just because doubling 5 produced 10.

Limit: A large reported value need not mean the true **branch** ran; **branch** choice depends on the earlier value.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Include equality examples when reviewing sorter **condition**s, and trace from original **input** through **output**. Draw your **input**, deciding step, result and limit.

Supported question: Which **input** detects the difference between the two written **condition**s? Explain its entire run.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.